

Differences among total and in vitro digestible phosphorus content of plant foods and beverages.

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OBJECTIVE: Among plant foods, grain products, legumes, and seeds are important sources of phosphorus (P). Current data on P content and absorbability of P from these foods are lacking. Measurement of in vitro digestible P (DP) content of foods may reflect absorbability of P. The objective of this study was to measure both total phosphorus (TP) and DP contents of selected foods and to compare the amounts of TP and DP and the proportion of DP to TP among different foods. METHODS: TP and DP content of 21 foods and drinks of plant origin were measured by inductively coupled plasma optical emission spectrometry. In DP analysis, samples were digested enzymatically in principle in the same way as in the alimentary canal before P analyses. The most popular national brands were chosen for analysis. RESULTS: The highest amount of TP (667 mg/100 g) was found in sesame seeds with hull, which also had the lowest percentage of DP (6%) to TP. Instead, in cola drinks and beer, the percentage of DP to TP was 87 to 100% (13 to 22 mg/100 g). In cereal products, the highest TP content (216 mg/100 g) and DP proportion (100%) were present in industrial muffins, which contain sodium phosphate as a leavening agent. Legumes contained an average DP content of 83 mg/100 g (38% of TP). CONCLUSION: Absorbability of P may differ substantially among different plant foods. Despite high TP content, legumes may be a relatively poor P source. In foods containing phosphate additives, the proportion of DP is high, which supports previous conclusions of the effective absorbability of P from P additives. Copyright © 2012 National Kidney Foundation, Inc. Published by Elsevier Inc. All rights reserved.